
Chapter 5. Joint Probability Distributions and Random Sample

Chapter 5: Joint Probability Distributions and Random Sample

- **5.1. Jointly Distributed Random Variables**
- **5.2. Expected Values, Covariance, and Correlation**
- **5. 3. Statistics and Their Distributions**
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Introduction

In chapter 3 and 4, we studied probability models for a **single random variable**.

However, many problem involve **several random variables simultaneously**

We first discuss probability models for the **joint behavior of several random variable**, putting special emphasis on the case in which **the variables are independent** of one another.

We then study **expected value of functions** of several random variables, including **covariance** and **correlation**

5.1. Jointly Distributed Random Variables

■ **The Joint Probability Mass Function for Two Discrete Random Variables**

Let X and Y be two discrete random variables defined on the sample space S of an experiment. The joint probability mass function $p(x,y)$ is defined for each pair of numbers (x,y) by

$$p(x, y) = P(X = x \text{ and } Y = y)$$

5.1. Jointly Distributed Random Variables

- Let A be any set consisting of pairs of (x,y) values. Then the probability $P[(X,Y) \in A]$ is obtained by summing the joint pmf over pairs in A :

$$p[(X,Y) \in A] = \sum_{(x,y) \in A} \sum p(x,y)$$

- Two requirements for a pmf

$$p(x,y) \geq 0 \quad \sum_x \sum_y p(x,y) = 1$$

■ Example 5.1

A large insurance agency services a number of customers who have purchased both a homeowner's policy and an automobile policy from the agency. For each type of policy, a deductible amount must be specified. For an **automobile policy**, the choices are \$100 and \$250, whereas for a **homeowner's policy** the choices are 0, \$100, and \$200.

Suppose an individual with both types of policy is selected at random from the agency's files. **Let X = the deductible amount on the auto policy, Y = the deductible amount on the homeowner's policy**

Suppose the joint pmf is given the accompanying **Joint Probability Table**

		y		
$p(x,y)$		0	100	200
x	100	0.20	0.10	0.20
	250	0.05	0.15	0.30

Then $p(100,100)=?$

$P(Y \geq 100)=?$

5.1. Jointly Distributed Random Variables

■ Example 5.1 (Cont')

$p(x,y)$		y		
		0	100	200
x	100	0.20	0.10	0.20
	250	0.05	0.15	0.30

$p(100,100) = P(X=100 \text{ and } Y=100) = 0.10$

$P(Y \geq 100) = p(100,100) + p(250,100) + p(100,200) + p(250,200) = 0.75$

5.1. Jointly Distributed Random Variables

■ The marginal probability mass function

The marginal probability mass functions of X and Y , denoted by $p_X(x)$ and $p_Y(y)$, respectively, are given by

$$p_X(x) = \sum_y p(x, y); \quad p_Y(y) = \sum_x p(x, y)$$

	Y_1	Y_2	$\dots$	Y_{m-1}	Y_m
X_1	$p_{1,1}$	$p_{1,2}$		$p_{1,m-1}$	$p_{1,m}$
X_2	$p_{2,1}$	$p_{2,2}$		$p_{2,m-1}$	$p_{2,m}$
$\dots$					
X_{n-1}	$p_{n-1,1}$	$p_{n-1,2}$		$p_{n-1,m-1}$	$p_{n-1,m}$
X_n	$p_{n,1}$	$p_{n,2}$		$p_{n,m-1}$	$p_{n,m}$

5.1. Jointly Distributed Random Variables

■ Example 5.2 (Ex. 51. Cont')

The possible X values are $x=100$ and $x=250$, so **computing row totals** in the joint probability table yields

		y		
$p(x,y)$		0	100	200
x	100	0.20	0.10	0.20
	250	0.05	0.15	0.30

$$p_x(100)=p(100,0)+p(100,100)+p(100,200)=0.5$$

$$p_x(250)=p(250,0)+p(250,100)+p(250,200)=0.5$$

$$p_x(x)=\begin{cases} 0.5, & x=100, 250 \\ 0, & \text{otherwise} \end{cases}$$

5.1. Jointly Distributed Random Variables

■ Example 5.2 (Cont')

$p(x,y)$		y		
		0	100	200
x	100	0.20	0.10	0.20
	250	0.05	0.15	0.30

$$p_y(0) = p(100,0) + p(250,0) = 0.2 + 0.05 = 0.25$$

$$p_y(100) = p(100,100) + p(250,100) = 0.1 + 0.15 = 0.25$$

$$p_y(200) = p(100,200) + p(250,200) = 0.2 + 0.3 = 0.5$$

$$p_Y(y) = \begin{cases} 0.25, & y = 0, 100 \\ 0.5, & y = 200 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} P(Y \geq 100) &= p(100,100) + p(250,100) + p(100,200) + p(250,200) \\ &= p_Y(100) + p_Y(200) = 0.75 \end{aligned}$$

5.1. Jointly Distributed Random Variables

■ The Joint Probability Density Function for Two Continuous Random Variables

Let X and Y be two **continuous random variables**. Then $f(x,y)$ is the joint probability density function for X and Y if for any two-dimensional set A

$$P[(X,Y) \in A] = \iint_A f(x,y) dx dy$$

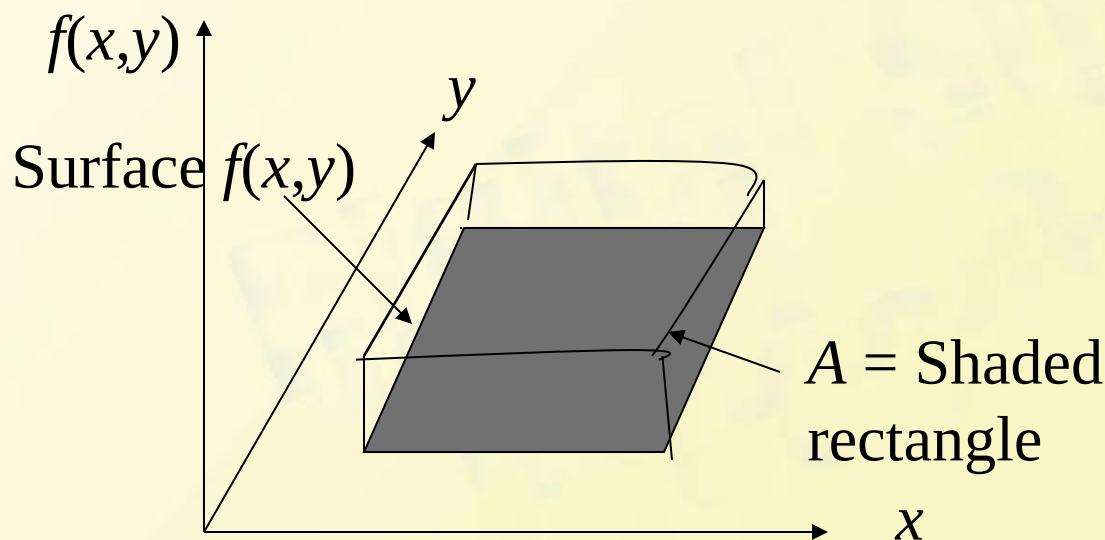
Two requirements for a joint pdf

1. $f(x,y) \geq 0$; for all pairs (x,y) in $\mathbb{R}^2$
2. $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x,y) dx dy = 1$

5.1. Jointly Distributed Random Variables

- In particular, if A is the **two-dimensional rectangle** $\{(x,y): a \leq x \leq b, c \leq y \leq d\}$, then

$$P[(X,Y) \in A] = P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x,y) dy dx$$



$P[(X,Y) \in A]$ = volume under density surface above A

5.1. Jointly Distributed Random Variables

■ Example 5.3

A bank operates both a drive-up facility and a walk-up window. On a randomly selected day, let X = the proportion of time that the drive-up facility is in use, Y = the proportion of time that the walk-up window is in use. Let the joint pdf of (X,Y) be

$$f(x,y) = \begin{cases} \frac{6}{5}(x + y^2) & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Verify that $f(x,y)$ is a joint probability density function;
2. Determine the probability $P(0 \leq X \leq \frac{1}{4}, 0 \leq Y \leq \frac{1}{4})$

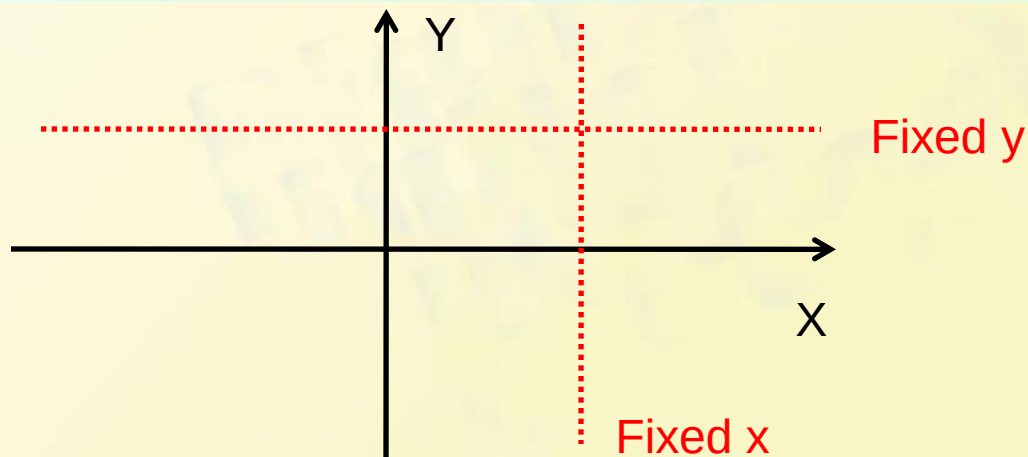
5.1. Jointly Distributed Random Variables

■ Marginal Probability density function

The marginal probability density functions of X and Y , denoted by $f_X(x)$ and $f_Y(y)$, respectively, are given by

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy \quad \text{for } -\infty < x < +\infty$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx \quad \text{for } -\infty < y < +\infty$$



5.1. Jointly Distributed Random Variables

■ Example 5.4 (Ex. 5.3 Cont')

The **marginal pdf of X**, which gives the probability distribution of busy time for the drive-up facility without reference to the walk-up window, is

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \int_0^1 \frac{6}{5} (x + y^2) dy = \frac{6}{5} x + \frac{2}{5}$$

for x in $(0,1)$; and 0 for otherwise.

The marginal pdf of Y is

$$f_Y(y) = \begin{cases} \frac{6}{5} y^2 + \frac{3}{5} & 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Then

$$P\left(\frac{1}{4} \leq Y \leq \frac{3}{4}\right) = \int_{1/4}^{3/4} f_Y(y) dy = 0.4625$$

■ Example 5.5

A nut company markets cans of deluxe mixed nuts containing **almonds, cashews, and peanuts**. Suppose the net weight of each can is **exactly 1 lb**, but the weight contribution of each type of nut is random. Because the three weights sum to 1, a joint probability model for any two gives all necessary information about the weight of the third type. **Let X = the weight of almonds in a selected can and Y = the weight of cashews. The joint pdf for (X,Y) is**

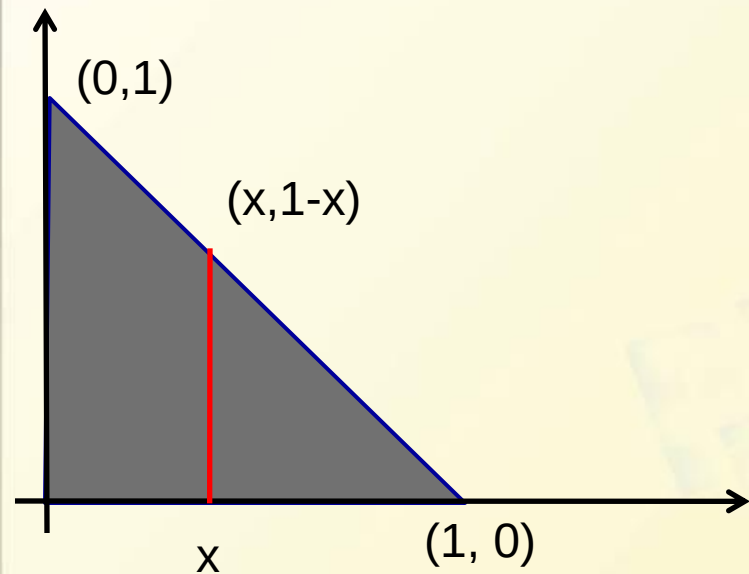
$$f(x,y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x+y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Verify that $f(x,y)$ is a joint probability density function;
2. Let $A = \{(x,y): 0 \leq x \leq 1, 0 \leq y \leq 1, \text{ and } x+y \leq 0.5\}$, $P(X,Y) \in A = ?$
3. $f_x(x) = ?$ And $f_y(y) = ?$

5.1. Jointly Distributed Random Variables

■ Solution:

$$f(x, y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



$$1: f(x, y) \geq 0$$

$$2: \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dy dx = \int_D \int f(x, y) dy dx$$

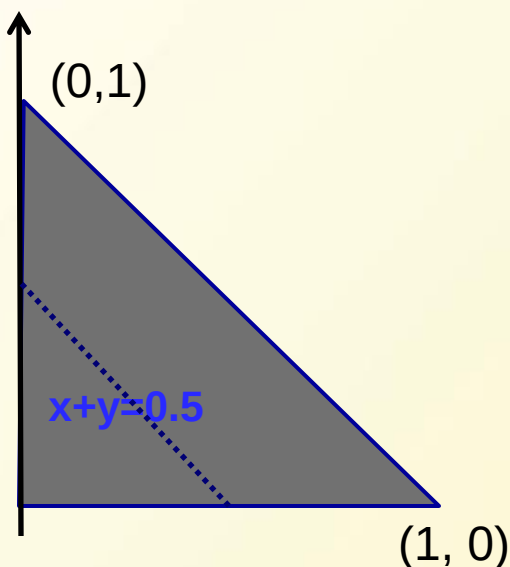
$$= \int_0^1 \left\{ \int_0^{1-x} (24xy) dy \right\} dx$$

$$= \int_0^1 12x(1-x)^2 dx = 1$$

5.1. Jointly Distributed Random Variables

■ Example 5.5 (Cont')

Let the two type of nuts together make up **at most 50%** of the can, then $A = \{(x, y); 0 \leq x \leq 1; 0 \leq y \leq 1, x + y \leq 0.5\}$

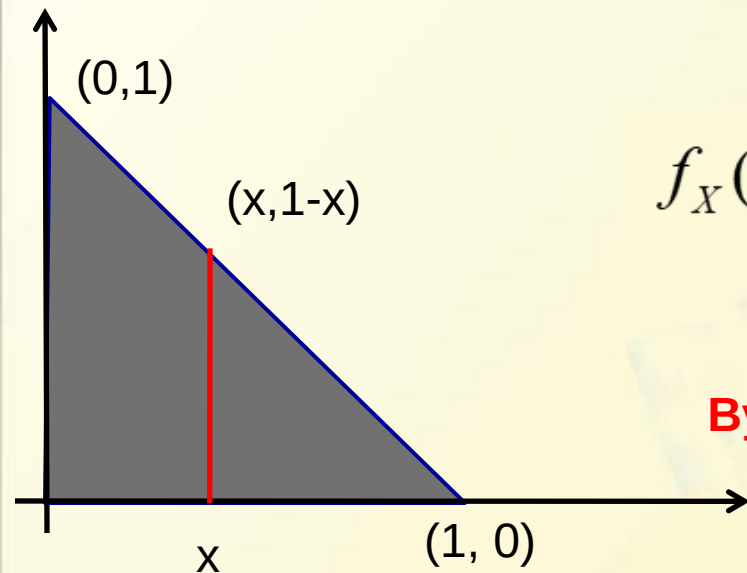


$$\begin{aligned} P((X, Y) \in A) &= \int \int_A f(x, y) dy dx \\ &= \int_0^{0.5} \left\{ \int_0^{0.5-x} (24xy) dy \right\} dx \\ &= 0.0625 \end{aligned}$$

5.1. Jointly Distributed Random Variables

■ Example 5.5 (Cont')

The **marginal pdf** for almonds is obtained by holding X fixed at x and integrating $f(x,y)$ along the vertical line through x :



$$f_X(x) = \begin{cases} \int_0^{1-x} (24xy) dy = 12x(1-x)^2, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

By symmetry

$$f_Y(y) = \int_0^{1-y} (24xy) dx = 12y(1-y)^2$$

5.1. Jointly Distributed Random Variables

■ Independent Random Variables

Two random variables X and Y are said to be independent if for **every pair of x and y values**,

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \text{when } X \text{ and } Y \text{ are discrete}$$

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{when } X \text{ and } Y \text{ are continuous}$$

Otherwise, X and Y are said to be dependent.

Namely, two variables are independent if their joint pmf or pdf is the product of the two marginal pmf's or pdf's.

5.1. Jointly Distributed Random Variables

■ Example 5.6

In the insurance situation of Example 5.1 and 5.2

$$p(100,100) = 0.1 \neq (0.5)(0.25) = p_X(100)p_Y(100)$$

		y		
$p(x,y)$		0	100	200
x	100	0.20	0.10	0.20
	250	0.05	0.15	0.30

So, X and Y are not independent.

5.1. Jointly Distributed Random Variables

■ Example 5.7 (Ex. 5.5 Cont')

Because $f(x,y)$ has the form of a product, X and Y would appear to be independent. However, although

$$f_X(x) = \int_0^{1-x} (24xy)dy = 12x(1-x)^2$$

$$f_Y(y) = \int_0^{1-y} (24xy)dx = 12y(1-y)^2 \quad \text{By symmetry}$$

$$f_x\left(\frac{3}{4}\right) = f_y\left(\frac{3}{4}\right) = \frac{9}{16}, \quad f\left(\frac{3}{4}, \frac{3}{4}\right) = 0 \neq \frac{9}{16} \cdot \frac{9}{16}$$

So, X and Y are not independent.

5.1. Jointly Distributed Random Variables

■ Example 5.8

Suppose that the lifetimes of two components are **independent** of one another and that the first lifetime, X_1 , has an **exponential distribution with parameter λ_1** whereas the second, X_2 , has an exponential distribution with parameter λ_2 . **Then the joint pdf is**

$$f(x_1, x_2) = f_{X_1}(x_1) \cdot f_{X_2}(x_2) = \begin{cases} \lambda_1 \lambda_2 e^{-\lambda_1 x_1 - \lambda_2 x_2} & x_1 > 0, x_2 > 0 \\ 0 & \text{otherwise} \end{cases}$$

Let $\lambda_1 = 1/1000$ and $\lambda_2 = 1/1200$. So that the expected lifetimes are 1000 and 1200 hours, respectively. **The probability that both component lifetimes are at least 1500 hours is**

$$P(1500 \leq X_1, 1500 \leq X_2) = P(1500 \leq X_1)P(1500 \leq X_2)$$

5.1. Jointly Distributed Random Variables

■ More than Two Random Variables

If $X_1, X_2, \dots, X_n$ are **all discrete rv's**, the **joint pmf** of the variables is the function

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

If the variables are **continuous**, the **joint pdf** of $X_1, X_2, \dots, X_n$ is the function $f(x_1, x_2, \dots, x_n)$ such that for any **n** intervals $[a_1, b_1], \dots, [a_n, b_n]$,

$$P(a_1 \leq X_1 \leq b_1, \dots, a_n \leq X_n \leq b_n) = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_n \dots dx_1$$

5.1. Jointly Distributed Random Variables

■ Multinomial Experiment

An experiment consisting of **n independent and identical trials**, in which each trial can result in any one of **r possible outcomes**.

Let $p_i = P(\text{Outcome } i \text{ on any particular trial})$, and define random variables by **$X_i = \text{the number of trials resulting in outcome } i$** ($i=1, \dots, r$). The joint pmf of $X_1, \dots, X_r$ is called the **multinomial distribution**. The joint pmf of $X_1, \dots, X_r$ can be shown to be

$$p(x_1, \dots, x_r) = \begin{cases} \frac{n!}{(x_1!)(x_2!)\dots(x_r!)} p_1^{x_1} \dots p_r^{x_r}, & x_i = 0, 1, \dots \text{ with } x_1 + x_2 + \dots + x_r = n \\ 0 & \text{otherwise} \end{cases}$$

Note: the case $r=2$ gives the binomial distribution.

5.1. Jointly Distributed Random Variables

- Independent

Definition:

The random variables $X_1, X_2, \dots, X_n$ are said to be **independent** if for every subset $X_{i1}, X_{i2}, \dots, X_{ik}$ of the variable (each pair, each triple, and so on), **the joint pmf or pdf of the subset is equal to the product of the marginal pmf's or pdf's.**

5.1. Jointly Distributed Random Variables

■ Example 5.10

When a certain method is used to collect a fixed volume of rock samples in a region, there are four resulting rock types. Let X_1 , X_2 , and X_3 denote the proportion by volume of rock types 1, 2 and 3 in a randomly selected sample. If the joint pdf of X_1, X_2 and X_3 is

$$f(x_1, x_2, x_3) = \begin{cases} kx_1x_2(1-x_3), & 0 \leq x_1 \leq 1, 0 \leq x_2 \leq 1, 0 \leq x_3 \leq 1, x_1 + x_2 + x_3 \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(1) Determine k

(2) $P(X_1 + X_2 \leq 0.5) = ?$

Solution:

(1) The k is determined by

$$\int \int \int_{D_1} f(x_1, x_2, x_3) = 1, D_1 : -\infty \leq x_i \leq \infty, i = 1, 2, 3 \quad \rightarrow \quad k=144.$$

(2) The probability that rocks of types 1 and 2 together account for **at most 50%** of the sample is

$$\int \int \int_{D_2} f(x_1, x_2, x_3) = 0.6066, D_2 : X_1 + X_2 \leq 0.5$$

5.1. Jointly Distributed Random Variables

■ Example 5.11

If $X_1, \dots, X_n$ represent the lifetime of **n components**, the components operate **independently of one another**, and each lifetime is **exponentially distributed** with parameter λ , then

(A) Joint pdf is?

(B) If there **n components** constitute a system that **will fail as soon as a single component fails**, then the probability that the system **lasts past t time is** ?

5.1. Jointly Distributed Random Variables

■ Solution:

$$\begin{aligned} \text{(A)} \quad f(x_1, x_2, \dots, x_n) &= (\lambda e^{-\lambda x_1})(\lambda e^{-\lambda x_2}) \dots (\lambda e^{-\lambda x_n}) \\ &= \begin{cases} \lambda^n e^{-\lambda \sum x_i}, & x_1 \geq 0; x_2 \geq 0; \dots, x_n \geq 0; \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

$$\begin{aligned} \text{(B)} \quad P(X_1 > t, X_2 > t, \dots, X_n > t) &= \int_t^\infty \dots \int_t^\infty f(x_1, x_2, \dots, x_n) dx_1 \dots dx_n \\ &= \left(\int_t^\infty \lambda e^{-\lambda x_1} dx_1 \right) \dots \left(\int_t^\infty \lambda e^{-\lambda x_n} dx_n \right) = e^{-n\lambda t} \end{aligned}$$

therefore,

$$P(\text{system lifetime} \leq t) = 1 - e^{-n\lambda t}, \text{ for } t \geq 0$$

5.1. Jointly Distributed Random Variables

■ Conditional Distribution

Let X and Y be two continuous rv's with joint pdf $f(x,y)$ and marginal X pdf $f_X(x)$. Then for any X values x for which $f_X(x) > 0$, the conditional probability density function of Y given that $X=x$ is

$$f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)}, -\infty < y < \infty$$

If X and Y are discrete, then

$$f_{Y|X}(y | x) = \frac{p(x, y)}{p_X(x)}, -\infty < y < \infty$$

is the conditional probability mass function of Y when $X=x$.

5.1. Jointly Distributed Random Variables

■ Example 5.12 (Ex.5.3 and Ex.5.4 Cont')

X = the proportion of time that a bank's drive-up facility is busy
and Y = the analogous proportion for the walk-up window.

(A) The conditional pdf of Y given that $X=0.8$ is ?

(B) The probability that the walk-up facility is busy **at most half the time given that $X=0.8$** is ?

Solution:

(A)

$$f_{Y|X}(y | 0.8) = \frac{f(0.8, y)}{f_X(0.8)} = \frac{1.2(0.8 + y^2)}{1.2(0.8) + 0.4} = \frac{1}{34}(24 + 30y^2), 0 < y < 1$$

(B)

$$f_{Y|X}(y \leq 0.5 | X = 0.8) = \int_{-\infty}^{0.5} f_{Y|X}(y | 0.8) dy = \int_{-\infty}^{0.5} \frac{1}{34}(24 + 30y^2) dy = 0.39$$